

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 3.2: Growth and Development in Animals — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 3.2: Growth and Development in Animals
Driving Question	DRIVING QUESTION: How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Empty Skin on the Maize Leaf: Two Ways Insects Grow	We observed an empty insect skin (exuvia) with a split on its back beside a larger insect; grasshopper nymphs that look like small adults with no pupa; housefly larvae that look nothing like the adult and a motionless pupa stage.	Insects follow two developmental pathways: incomplete metamorphosis (egg, nymph, adult: 3 stages) as in the grasshopper, and complete metamorphosis (egg, larva, pupa, adult: 4 stages) as in the housefly and butterfly.	We can explain that the two pathways differ in the number of stages and in whether the young resemble the adult, but we cannot yet explain how the insect got bigger than its old skin or why grasshoppers grow in sudden jumps rather than smoothly.
Lesson 2 Stepwise Growth: Why Hard-Shelled Animals Grow in Jumps	We measured and plotted the length and mass of insect nymphs over time and saw that the graph rose in steps: flat periods with no length change, followed by sudden jumps when the insect moulted, leaving an empty exuvia.	Animals with a hard exoskeleton cannot grow smoothly because the shell does not stretch. They must moult (ecdysis) first, then swell while the new cuticle is soft, giving an intermittent (stepwise) growth curve.	The grasshopper on the maize leaf is bigger than its exuvia because it split the old shell, expanded its body before the new cuticle hardened, and so grew in a sudden jump. New question: how do animals with internal skeletons grow compared to insects?
Lesson 3 Growth Curves: Why Do Vertebrates and Soft-Bodied Animals Grow Smoothly in an S-Shape While Insects with Hard Shells Grow in Steps?	We plotted mass and height data for a Kenyan broiler chick, a child, and a grasshopper nymph. The chick and child graphs formed smooth S-shaped curves, while the grasshopper's length stayed flat and jumped at each moult.	Vertebrates and soft-bodied animals grow continuously in a sigmoid curve with lag, exponential, deceleration and stationary phases. Growth rate = (final value - initial value) / time interval, and it is highest in the exponential phase and near zero in the stationary phase. Hard-shelled insects show intermittent, stepwise growth because the exoskeleton cannot stretch.	The chick has no rigid shell, so cells can divide and increase size continuously: slow at first, then rapid, then slowing as it nears adult size set by genes and resources. The grasshopper must shed its exoskeleton (leaving the exuvia) before it can grow, so its size increases in steps. What triggers and controls the timing of moulting and metamorphosis is still unanswered and is the focus of Lesson 4.
Lesson 4	We analysed flowcharts of experiments on	Brain Hormone signals the prothoracic	The grasshopper or cicada outgrows its

Hormones as the Control Switches of Moulting and Metamorphosis	larvae: with the brain/neurosecretory cells removed the larva did not moult; with Ecdysone injected it moulted again; with extra Juvenile Hormone it moulted into another larva instead of a pupa; with the corpora allata removed early it formed a small pupa early. We also compared these with the empty exuvia and the larger grasshopper or cicada from the school garden.	glands to release Ecdysone, which triggers moulting. Juvenile Hormone decides the kind of moult: high levels keep the larval form, low levels allow pupation, and none allows adult emergence. Hormones interact like a control system, so moulting and shape change are regulated and not random.	hard shell because hormones trigger a new, larger cuticle to form beneath the old one, the old one splits and is left as the exuvia, and the insect expands before the new cuticle hardens, so growth happens in steps. Insects with complete metamorphosis, like caterpillars, change shape completely because Juvenile Hormone falls and allows pupation, while grasshopper nymphs keep JH until the last moult and so look like small adults.
Lesson 5 The Chemical Switch: How Juvenile Hormone Decides Whether an Insect Stays Young or Becomes an Adult	We viewed animations and graphs showing ecdysone peaks before every moult, and JH levels that stay high during nymph/larval moults but fall before the pupa and adult stages. We also saw that grasshoppers go from nymph to adult without a pupa, while butterflies have a pupal stage. The exuvia in the garden is the old exoskeleton left after a moult.	Moulting is triggered by ecdysone, and the level of Juvenile Hormone decides what type of moult occurs: high JH means the insect moults into another nymph/larva, while falling JH leads to a pupa (complete metamorphosis) or an adult (incomplete metamorphosis). Growth is stepwise because the exoskeleton must be shed before the insect can get bigger. Metamorphosis is a controlled, hormone-regulated process, not 'melting without control'.	An insect outgrows its hard shell by releasing ecdysone, which causes it to shed its old exoskeleton and expand before the new one hardens. A grasshopper keeps a high-to-declining JH level through several nymph moults and becomes an adult without a pupa, while a butterfly caterpillar's JH falls dramatically before it forms a pupa and then an adult, which is why some animals change shape completely and others just grow bigger.
Lesson 6 Final Explanation: Controlled Growth, Metamorphosis and Survival in Kenyan Insects	We observed an empty exuvia with a split down its back beside a larger newly emerged grasshopper or cicada, caterpillars becoming pupae and butterflies, and insect growth curves with steps and flat sections. We also observed how our Day 1 models differed from our final models.	We learned that an insect's hard exoskeleton cannot stretch, so the insect moults (ecdysis) under hormone control: ecdysone triggers moulting and juvenile hormone determines whether the next stage stays juvenile or develops toward the adult. This produces intermittent growth. Incomplete metamorphosis (egg, nymph, adult) gives small adult-like nymphs, while complete metamorphosis (egg, larva, pupa, adult) gives very different life stages that use different foods and survive harsh conditions.	We can explain that an animal outgrows its hard shell by making a new, larger exoskeleton beneath the old one, then splitting and leaving the old exuvia while the new one is still soft and expandable. Controlled hormone-regulated growth and developmental transitions help animals optimize resource use, reduce competition between young and adults, and survive harsh environments. New question: How might human pesticides disrupt these exact hormonal pathways in agricultural pests?

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